
ACROPOLIS
Enlightening Wisdom
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Computer Science and Engineering
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PROBLEM STATEMENTS
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ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

1. Large-scale agricultural pest management systems relying on single or centralized units face limitations in coverage efficiency, scalability, response time, and resource optimization. There is a need for an autonomous, decentralized multi-agent framework capable of real-time hotspot identification and coordinated operation across extensive farmlands to improve precision, reduce chemical overuse, and enhance overall system reliability.

2. Urban emergency vehicles often face delays due to congested traffic and uncoordinated signal systems that operate on fixed or semi-adaptive timings. There is a need to design an intelligent, city-scale traffic coordination system that dynamically prioritizes emergency vehicles across multiple interconnected intersections using real-time data. The system should create a seamless green corridor for emergency vehicles while minimizing disruption to regular traffic flow and maintaining overall traffic stability. Teams may use simulation, mock data, or public datasets to demonstrate their solution.

3. Urban waste management suffers from poor segregation at the source, leading to contamination of recyclable materials and increased landfill burden. Current systems rely on manual processes, limited awareness, and lack real-time visibility into waste composition and available recycling infrastructure. There is a need for a smart, technology-driven system that enables accurate waste identification and segregation at the household level while connecting waste streams with appropriate recycling and processing facilities to improve efficiency and sustainability in urban waste management.

4. Current urban safety systems rely on fragmented forecasting and surveillance mechanisms, leading to reactive responses and delayed intervention. There is a need for an integrated, real-time framework that dynamically assesses evolving urban risks using multi-source data while ensuring ethical, transparent, and responsible governance.

WEB DEVELOPMENT

1. Citizens often struggle to access immediate legal guidance during disputes or urgent situations, leading to uninformed decisions and escalation of conflicts. At the same time, legal information is complex, fragmented, and difficult for the general public to understand or access in a timely manner. There is a need for an intelligent legal assistance system that can act as a real-time digital advocate by analyzing live dispute scenarios between parties and helping individuals understand their rights, responsibilities, and possible legal actions. Alongside this, the system should function as an accessible legal knowledge layer that simplifies complex laws and provides updates on evolving legislation.

2. The healthcare ecosystem operates in silos, where critical information about doctors, ICU beds, medical equipment, ambulances, and infrastructure is fragmented across hospitals and often managed through manual or disconnected systems. This lack of real-time visibility leads to delays in emergencies, resource underutilization, and inefficient patient transfers. There is a need for an intelligent, real-time healthcare resource sharing and coordination platform that enables dynamic visibility, allocation, and optimization of medical resources across multiple hospitals to reduce emergency delays and improve system efficiency.

3. In many residential neighborhoods, households are increasingly generating renewable energy at the local level. However, there is no efficient, transparent, and automated mechanism that enables direct exchange of surplus energy between nearby homes. Existing systems rely heavily on centralized intermediaries, limiting flexibility, fair value realization, and optimal local energy utilization. Design a solution that enables secure, peer-based energy transactions within a smart neighborhood ecosystem without depending on a central authority. The system may include a web-based platform that allows households to list surplus energy, discover nearby energy providers, monitor transactions, and manage energy exchange through interactive dashboards and APIs.

WEB DEVELOPMENT

4. Regional weather forecasts lack the spatial accuracy required to predict farm-level micro-climate variations, exposing smallholder farmers to unexpected localized climate events. There is a need for a system that delivers accurate, short-term, hyper-local climate predictions to support timely and informed agricultural decision-making. Design a solution that delivers farm-level climate insights through an interactive web platform where farmers can access location-based forecasts, visualize weather conditions through maps and dashboards, and receive alerts for potential climate risks affecting their farms.

CIVIL ENGINEERING

1. High-rise buildings experience continuous vibrations due to environmental and operational factors, but this data is not intelligently analyzed in real time to detect early structural issues. The lack of AI-driven anomaly detection increases the risk of undetected damage and delayed maintenance. There is a need for a deep learning-based system that continuously monitors vibration data and provides early warning alerts. The system should integrate with structural health monitoring sensors installed in buildings and provide dashboards for engineers and facility managers to assess structural stability, detect anomalies, and support preventive maintenance decisions in civil infrastructure.

2. Construction sites often experience accidents due to unsafe worker behavior and non-compliance with safety protocols. Manual supervision is insufficient to continuously monitor large and dynamic construction environments. There is a need for an AI-powered safety monitoring system that analyzes live site footage to detect safety violations, identify potential risk situations, and provide real-time alerts to supervisors. The system should support centralized monitoring of multiple construction sites and provide safety analytics dashboards for project managers to track compliance with construction safety standards and improve overall site safety management.

3. Static BIM models are frequently "data graveyards" once construction ends. Currently, maintenance schedules do not auto-generate from these models, forcing facility managers to manually input thousands of assets into disconnected spreadsheets. This gap leads to reactive "run-to-failure" maintenance, inaccurate lifecycle cost forecasting, and a total loss of the "Digital Twin" advantage. Design a system that can analyze a 3D BIM model of a building and generate predictive maintenance schedules, lifecycle cost projections, and structural component degradation insights, helping facility managers make informed long-term maintenance and infrastructure management decisions

BLOCKCHAIN

1. Farmers practicing sustainable agriculture face challenges in accurately measuring and verifying carbon sequestration, limiting their ability to access and monetize carbon credits. There is a need for an automated, scalable system that integrates remote sensing and ground-based data to enable reliable carbon quantification and transparent credit verification. Design a solution that leverages blockchain-based registries and smart contracts to record carbon sequestration data, verify carbon credits, and ensure transparent, tamper-proof tracking of carbon credit generation and transactions. The system may integrate remote sensing and field data sources,
while using blockchain to provide trust, traceability, and decentralized verification for carbon credit issuance and trading.

2. MSMEs often face severe cash flow challenges due to delayed invoice payments from larger buyers. Traditional financing options are slow, collateral-heavy, and inaccessible for small businesses.
Build a decentralized invoice financing marketplace where MSMEs tokenize unpaid invoices and receive immediate liquidity from investors, while smart contracts manage settlements .

3. Disaster relief funds often suffer from delays, leakage, and lack of transparency. Beneficiaries may not receive aid on time, and donors struggle to verify how funds are used. Design a stablecoin-based relief distribution system that enables rapid, trackable fund transfers directly to verified beneficiaries, with optional controls on how funds can be spent.

CYBER SECURITY

1. The danger of fake news isn't just the lie itself; it’s the coordinated amplification. Currently, systems focus mainly on fact checking, labeling posts as false after they spread, but fail to identify where the narrative originated and how it propagated across networks. Without tracing the initial source ("patient zero") and identifying the key amplification accounts ("super-spreaders"), mitigation efforts only address the symptoms rather than the root cause. Design a solution that can trace the origin of misinformation, analyze propagation patterns across platforms, and identify coordinated amplification networks. The system should incorporate cybersecurity techniques such as bot detection, anomaly detection, and network threat intelligence analysis to identify malicious actors, coordinated bot campaigns, and suspicious amplification clusters, enabling platforms and authorities to respond proactively to large-scale misinformation threats.

2. Bug bounty programs are highly effective but consistently overwhelm security teams with a massive volume of submissions. A significant percentage of these reports are false positives, duplicates, or low-impact issues—often referred to as the "triage tax." Security engineers spend countless hours manually parsing reports, reproducing bugs, validating claims, and assessing severity. This manual bottleneck delays the patching of genuinely critical vulnerabilities, leaving organizations exposed to exploitation. Create a framework that can automatically validate, classify, and prioritize bug bounty vulnerability reports using AI-driven exploit likelihood scoring to reduce false positives and improve response time.

3. A highly scalable, edge-powered fintech security platform that enables multiple banks to collaboratively detect and prevent complex, multi-hop financial fraud and money laundering attacks without sharing raw customer data. The system should enable institutions to identify suspicious transaction patterns, detect coordinated fraud networks, and strengthen cross-bank threat intelligence, while maintaining strict compliance with data privacy regulations and financial security standards.

APP DEVELOPMENT

1. Despite the architectural and intellectual brilliance of ancient Indian universities like Nalanda, Takshashila, and Vikramshila, they remain mostly ruins or text-based history for the modern student. The challenge is to bridge this gap using Augmented Reality (AR) to teleport these "Lost Temples of Knowledge" into a user's current physical surroundings. Participants must build an AR experience that doesn't justshow a 3D model, but recreates the pedagogy (teaching style) and atmosphere ofancient India through immersive storytelling.

2. Create a system that digitizes ancestral medicinal knowledge while mitigating therisks of misinformation. Participants must go beyond simple image classification to provide 'Digital Masterclasses'—integrating 3D anatomical overlays or AR-guided tutorials that teach the safe, clinical preparation of traditional remedies. The goal is to transform a simple leaf scan into a structured, expert-verified educational experience.

3. Manual evaluation of handwritten government exam papers is time-consuming,
inconsistent, and prone to human bias. There is a need for a secure and standardized digital system that can support the efficient assessment of offline written answer sheets while ensuring fairness, transparency, and scalability in large-scale examination processes.

4. Emergency vehicles often face delays due to traffic congestion, road blockages, and rapidly changing road conditions. Conventional navigation tools rely mainly on shortest-path calculations and basic traffic updates, which are insufficient during time-critical emergency situations. There is a need to develop an intelligent emergency routing application that identifies optimal routes using real-time traffic data and adapts navigation during transit. Design a mobile application for emergency responders and dispatch centers that provides real-time navigation assistance, route reliability insights, and alerts about traffic disruptions. The app should allow live route monitoring, communication with dispatch teams, and continuous route updates based on real-time traffic and road condition data to support faster and more informed emergency response.

DOUBLE REWARD OPPORTUNITY

As part of **Prayatna 3.0** Hackathon, we're introducing an exclusive Robotics Side Quest where selected teams will work with the **SO-01 robotic arm** and train it to perform real-world tasks.

Using the Solo CLI, teams will experience the complete robotics ML pipeline:
Configure → Record Dataset → Train → Inference

Instead of writing traditional automation scripts, you will teach the robot through demonstrations and machine learning. Your model will learn from the dataset you record and execute the task autonomously during inference.

Repository: https://github.com/GetSoloTech/solo-cli

1. Smart Charging Assistant

Base Task
Train the robot to pick up a charger and plug it into a socket reliably.

- The robot should learn how to:
- Grasp the charger correctly
- Orient it in the right direction
- Align and insert it into the socket

The focus is on learning through demonstrations, so the robot should be able to repeat the task consistently after training.

Stand-Out Ideas (Optional)
The base task alone is enough to complete this problem statement. However, teams looking to differentiate their solution may layer additional interaction on top of their trained model.
For example, you could trigger the trained behaviour using voice commands, integrate vision to detect charger placement, or build a simple interface that lets users instruct the robot naturally.
These additions are not required, but they can help your project stand out.

DOUBLE REWARD OPPORTUNITY

2. Robot Game Master

Base Task

Train the robot to interact with a simple tabletop game by picking and placing game elements.

Examples could include:

- Placing pieces in Tic-Tac-Toe
- Moving objects in a small board game
- Interacting with elements in a Hot Wheels style setup

The robot should learn how to grasp, move, and place pieces accurately based on the demonstrations used during training.

Stand-Out Ideas (Optional)

Once the robot can perform the base interaction, teams may choose to extend the experience.

For instance, the robot could be triggered through voice interaction, connected to a simple game logic system, or integrated with a small vision pipeline that reads the board state.

Such additions are optional, but they can elevate the overall experience of the project.

DOUBLE REWARD OPPORTUNITY

3.Open Innovation

Base Task

Design your own robotic task and train the robot to perform it using the Solo training pipeline.

This could involve:

- Manipulating objects
- Performing a repetitive physical action
- Demonstrating a useful robotic capability

The emphasis should remain on robot learning through recorded demonstrations and trained models.

Stand-Out Ideas (Optional)

Teams are encouraged to bring creativity to this category. After implementing the base robotic behaviour, you may choose to enhance the interaction or presentation.

This could include adding voice triggers, a visual interface, or integrating the robot into a broader interactive system.

The goal is to showcase what a trained robotic model can enable, not just the task itself.